
How the six laboratories work

ORIGINAL ILLUSTRATIVE PLAY

These are original engine-assisted study continuations from this course's positions. They are not historical master games and no invented players or tournament results are assigned. The continuations reach roughly move twenty and stop for further practice.

THINK BEFORE REVEALING

At each checkpoint, name the opponent's threat, compare two candidates and calculate a forcing reply. Use a board to replay the sequence. The engine comparison supplies concrete evidence, while the opening explanation supplies a plan.

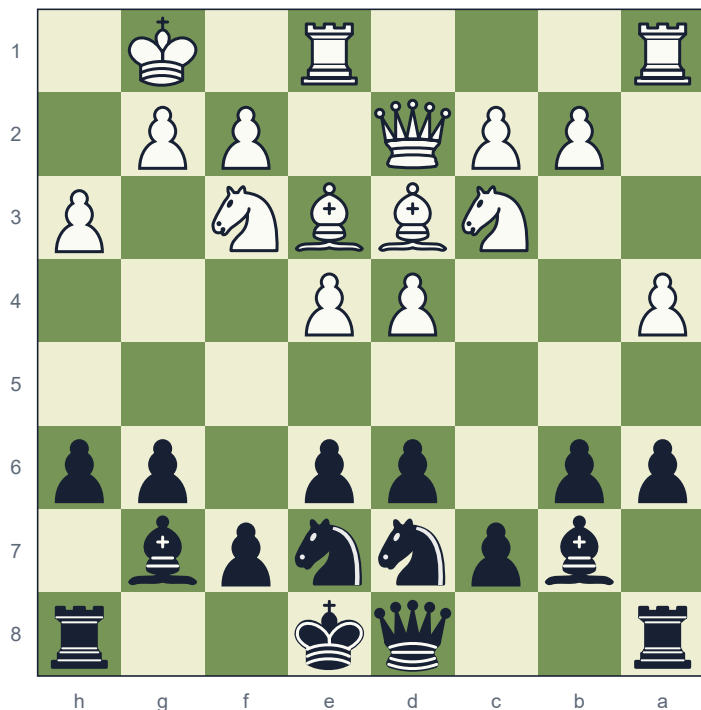
MORE THAN ONE GOOD MOVE

Close engine scores do not establish a unique answer. The finite search can prefer a different move when run longer. Accept another candidate if you can support it with a sound variation and a coherent explanation.

PRACTICE FROM BOTH SIDES

Play each starting position twice, switching colors. Record the first decision you could not explain and compare it with the branch lesson after the game.

HB-LAB-01 / The full formation as a study position



Laboratory start - Black to move

OPENING BACKGROUND

This illustrative route shows the recognizable formation when White develops quietly. The knights avoid blocking the fianchetto bishops, but Black has used many pawn tempi and is still uncastled.

READ THE CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king e8; central-file pawns d6, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

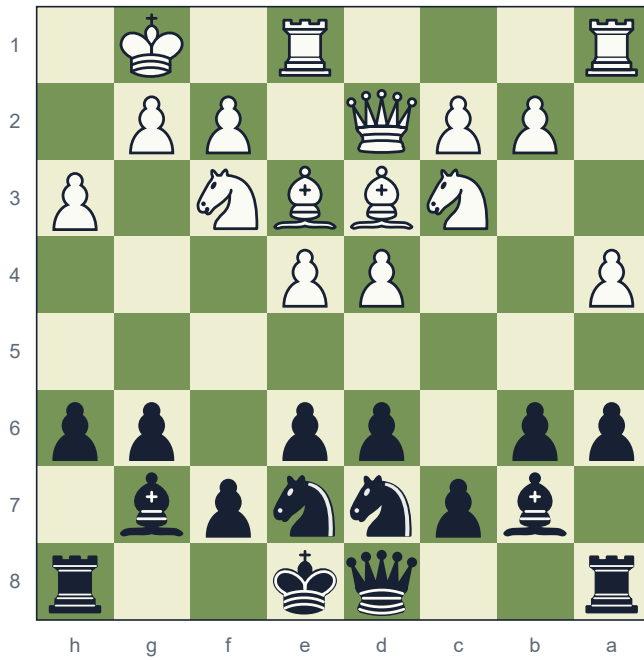
White owns more space and can prepare d5, e5 or a wing attack. Finishing the diagram is not a strategic achievement if Black then has no active move. Decide which parts still apply after the moves already played.

BEFORE PLAYING

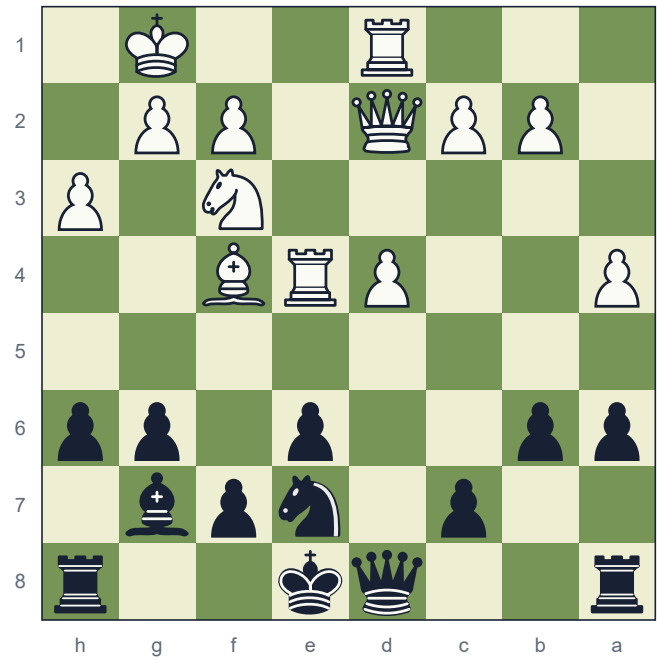
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-01 / play and explain 1

11... Nf6 12. Rad1 d5 13. Bf4 Nxe4 14. Nxe4 dxe4 15. Bxe4 Bxe4 16. Rxe4



Before this segment - Black to move



After this segment - Black to move

11... NF6

Develops the king knight toward e4 and d5. Its location also blocks some long-diagonal bishop attacks, but makes it a possible target of e5.

13... NXE4

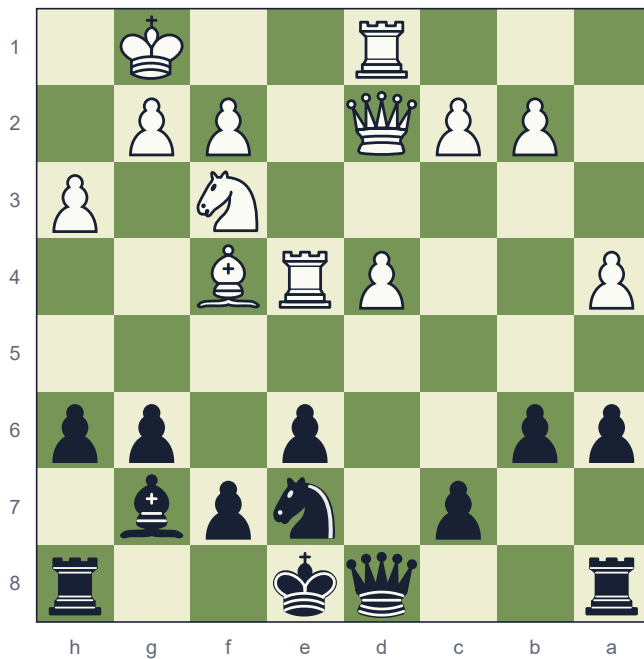
The knight captures a pawn on e4. Calculate the best reply before assuming a recapture, then recount material and opened lines.

REASSESS

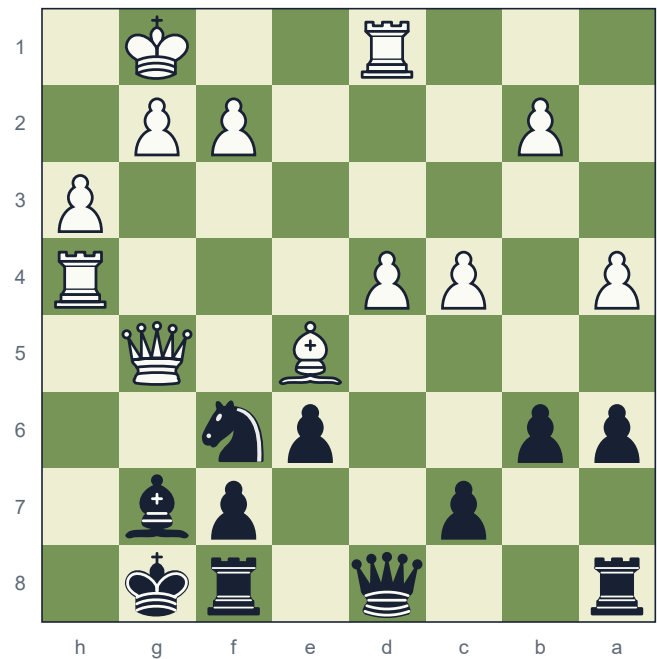
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-01 / play and explain 2

16... Nd5 17. Be5 O-O 18. c4 Nf6 19. Rh4 g5 20. Nxd5 hxd5 21. Qxd5



Before this segment - Black to move



After this segment - Black to move

16... ND5

Places the knight on d5, attacking bishop on f4. A legal capture may still be unsound if the target is defended.

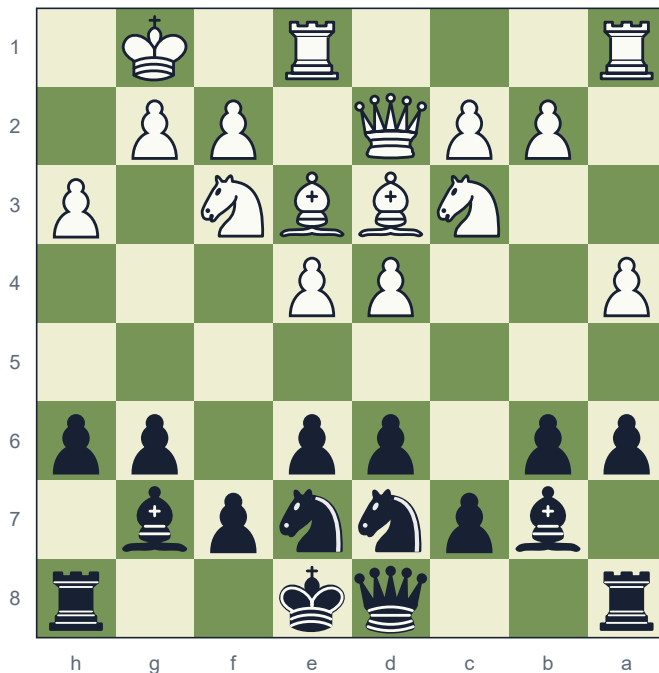
18... NF6

Places the knight on f6, attacking rook on e4. A legal capture may still be unsound if the target is defended.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-01 / decision 1



Black to move

COMPARE CANDIDATES

Nf6 and d5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king e8; central-file pawns d6, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NF6

11... Nf6 12. Rad1 Search estimate: -1.12 from Black's perspective.

CANDIDATE D5

11... d5 12. Bf4 g5 13. Bg3 O-O 14. e5 c5 15. h4 Search estimate: -1.16 from Black's perspective.

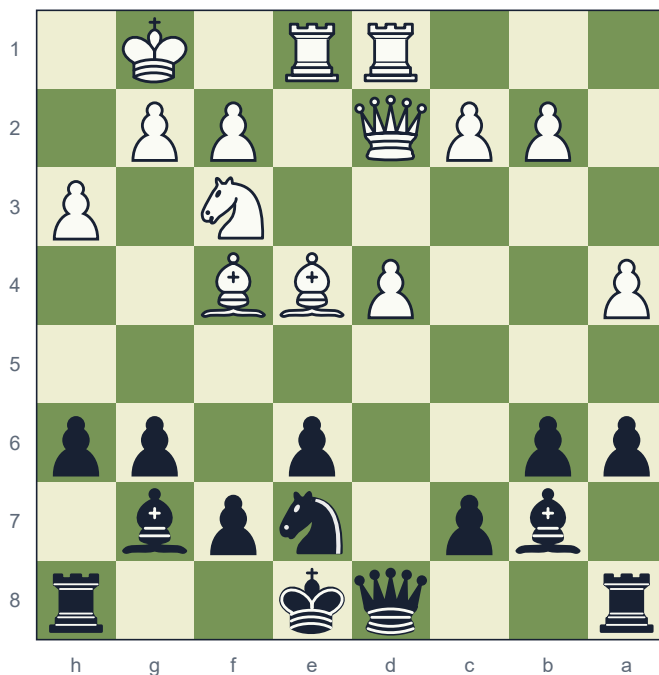
EXPLAIN THE PREFERRED MOVE

Develops the king knight toward e4 and d5. Its location also blocks some long-diagonal bishop attacks, but makes it a possible target of e5.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-01 / decision 2



Black to move

COMPARE CANDIDATES

Bxe4 and Bd5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4. Black king e8; central-file pawns e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BXE4

15... Bxe4 16. Rxe4 Nd5 17. Be5 Nf6 18. Rf4 g5 19. Bxf6 Search estimate: -1.21 from Black's perspective.

CANDIDATE BD5

15... Bd5 16. c4 Bxe4 17. Rxe4 c6 18. b3 b5 19. Re2 Search estimate: -1.41 from Black's perspective.

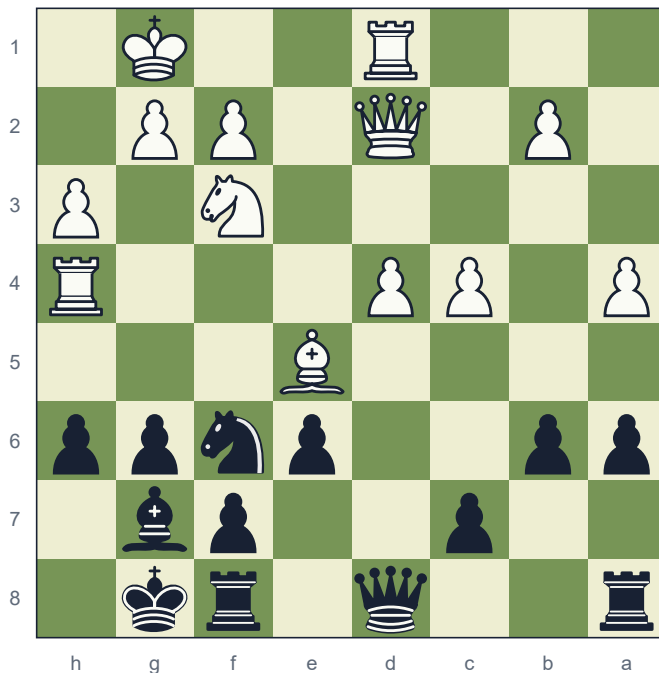
EXPLAIN THE PREFERRED MOVE

The bishop captures a bishop on e4. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-01 / decision 3



Black to move

COMPARE CANDIDATES

g5 and h5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king g8; central-file pawns e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE G5

19... g5 Search estimate: -0.95 from Black's perspective.

CANDIDATE H5

19... h5 20. Qc3 Nd7 21. Bxg7 Kxg7 22. Re1 Re8 23. Rhe4 Search estimate: -1.16 from Black's perspective.

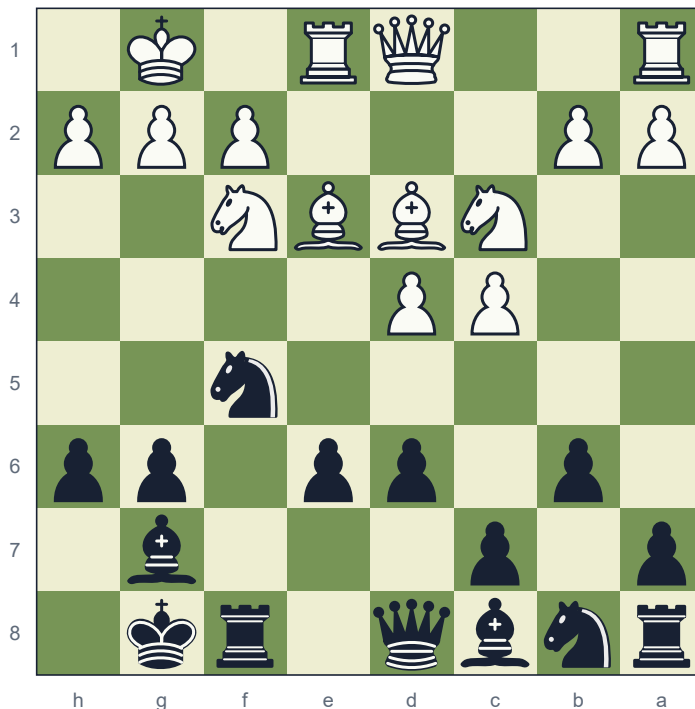
EXPLAIN THE PREFERRED MOVE

Places the pawn on g5, attacking rook on h4. A legal capture may still be unsound if the target is defended.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-02 / The three-pawn center



Laboratory start - Black to move

OPENING BACKGROUND

White occupies c4-d4-e4. Black chooses a partial formation and castles instead of insisting on both fianchettos before addressing the center.

READ THE CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king g8; central-file pawns d6, e6, c7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

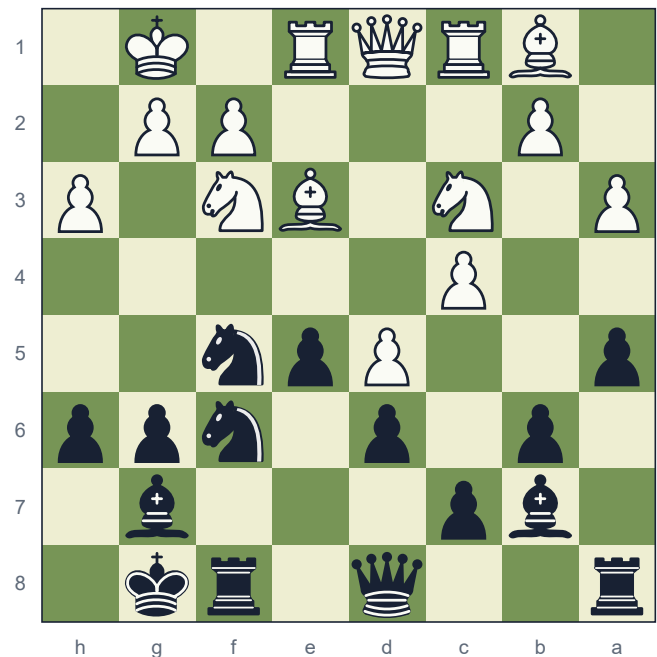
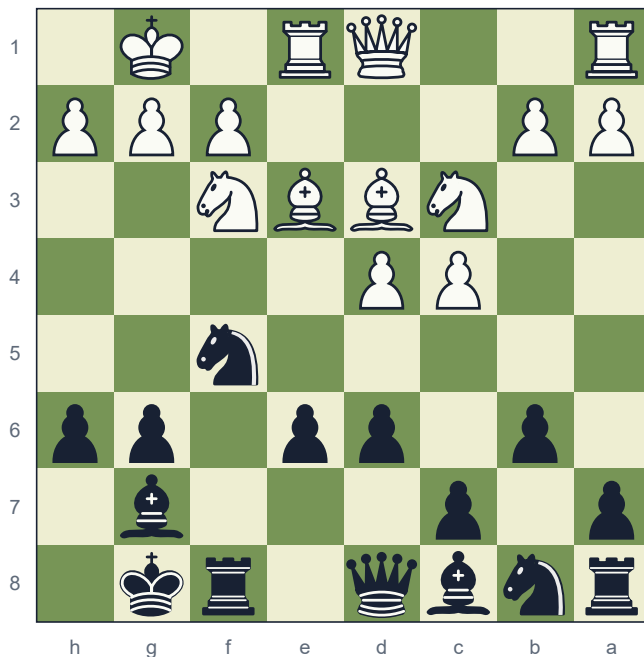
White can gain more space with d5 or build a kingside attack with f4. Do not add ...a6 and ...h6 automatically under a development deficit. Decide which parts still apply after the moves already played.

BEFORE PLAYING

Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-02 / play and explain 1

11... Nd7 12. Rc1 Bb7 13. d5 e5 14. Bb1 a5 15. a3 Nf6 16. h3



11... ND7

Develops a knight toward central supporting squares. Keep a useful onward route so the compact setup does not become overcrowded.

13... E5

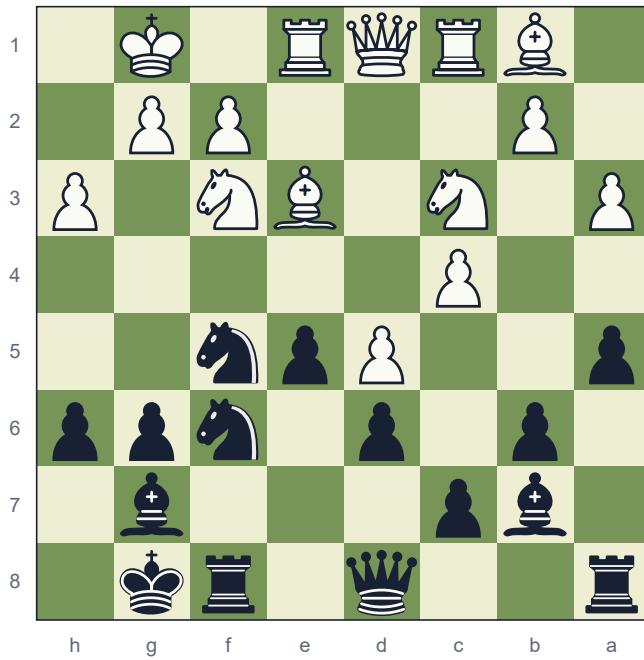
Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

REASSESS

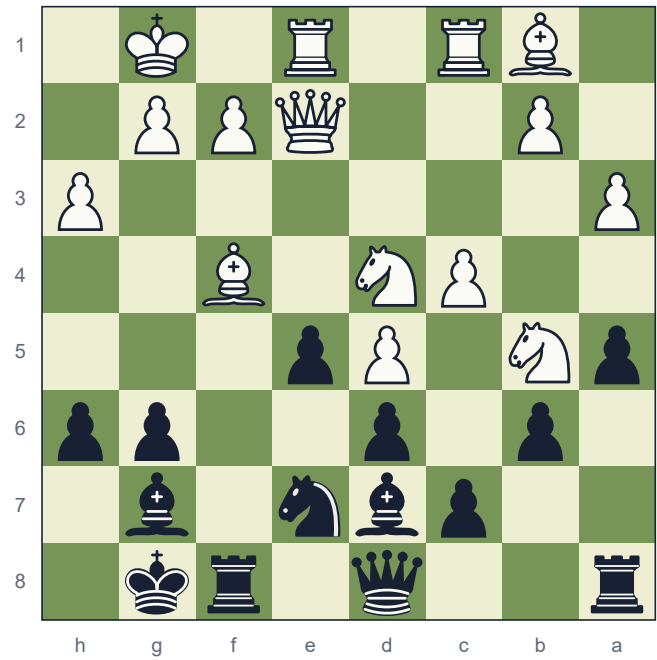
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-02 / play and explain 2

16... Bc8 17. Bd2 Bd7 18. Qe2 Nh5 19. Nb5 Ne7 20. Nfd4 Nf4 21. Bxf4



Before this segment - Black to move



After this segment - Black to move

16... BC8

Places the bishop on c8. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

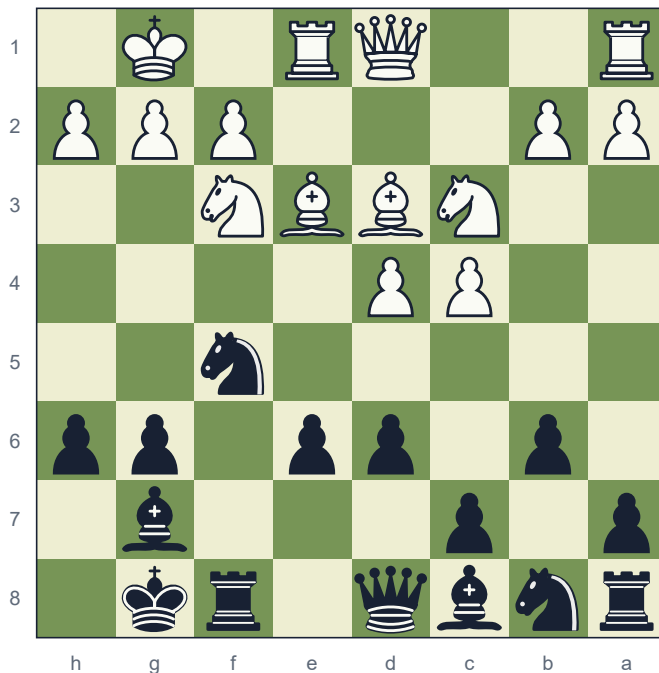
18... NH5

Places the knight on h5. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-02 / decision 1



Black to move

COMPARE CANDIDATES

Nd7 and Nc6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king g8; central-file pawns d6, e6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE ND7

11... Nd7 12. Be4 Rb8 13. h3 a6 14. Qd2 g5 15. Rac1 Search estimate: -0.76 from Black's perspective.

CANDIDATE NC6

11... Nc6 12. Be4 Bb7 13. d5 Ne5 14. Nd4 Nxe3 15. Rxe3 Search estimate: -0.80 from Black's perspective.

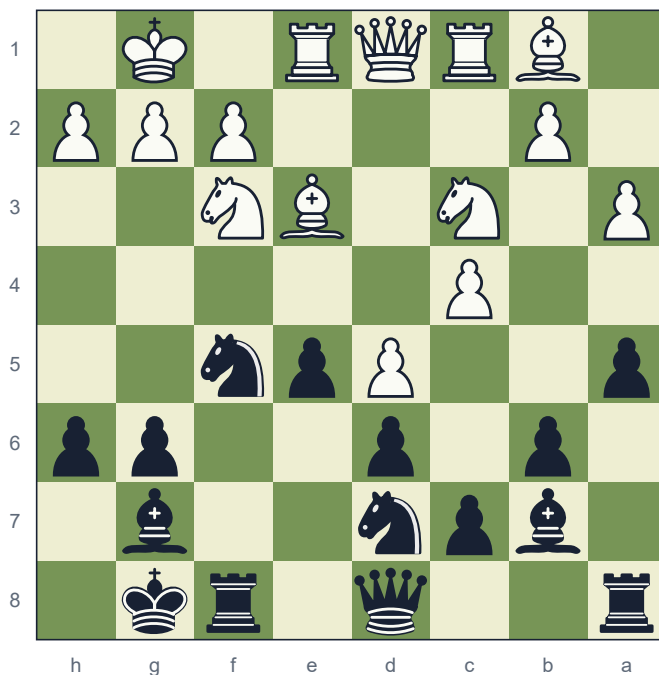
EXPLAIN THE PREFERRED MOVE

Develops a knight toward central supporting squares. Keep a useful onward route so the compact setup does not become overcrowded.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-02 / decision 2



Black to move

COMPARE CANDIDATES

Nf6 and Ba6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d5. Black king g8; central-file pawns e5, d6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NF6

15... Nf6 16. h3 Bc8 17. Bd2 Bd7 18. Nb5 Rc8 19. a4 Search estimate: -0.72 from Black's perspective.

CANDIDATE BA6

15... Ba6 16. b3 Search estimate: -0.77 from Black's perspective.

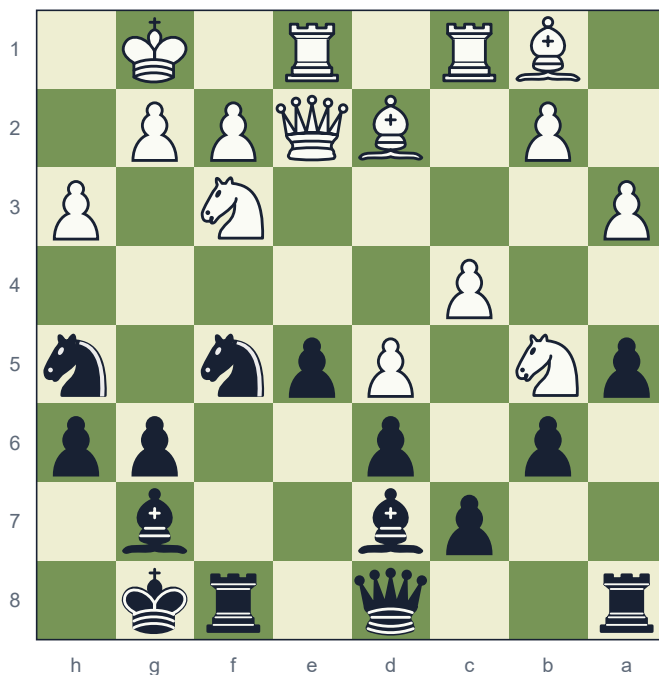
EXPLAIN THE PREFERRED MOVE

Develops the king knight toward e4 and d5. Its location also blocks some long-diagonal bishop attacks, but makes it a possible target of e5.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-02 / decision 3



Black to move

COMPARE CANDIDATES

Ne7 and Nf6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d5. Black king g8; central-file pawns e5, d6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NE7

19... Ne7 20. Rc3 Search estimate: -0.66 from Black's perspective.

CANDIDATE NF6

19... Nf6 20. g4 Ne7 21. Nh4 g5 22. Ng6 Nxd6 23. Bxd6 Search estimate: -1.09 from Black's perspective.

EXPLAIN THE PREFERRED MOVE

Develops the king knight while keeping the f-pawn flexible. It may support a central break or move toward g6 or c6 if safe.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-03 / An immediate h-pawn attack



Laboratory start - Black to move

READ THE CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king g8; central-file pawns c6, d6, e7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

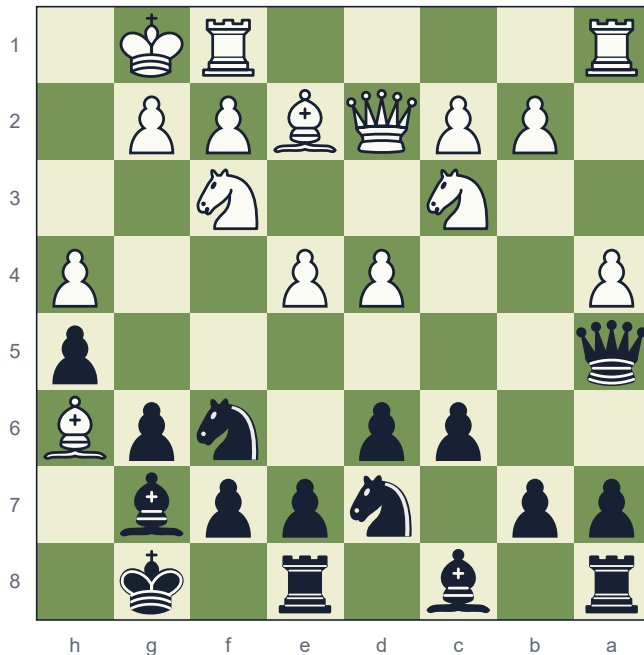
g4 can attack the new h5 hook. A blocked h-pawn does not mean the king side is permanently safe. Decide which parts still apply after the moves already played.

BEFORE PLAYING

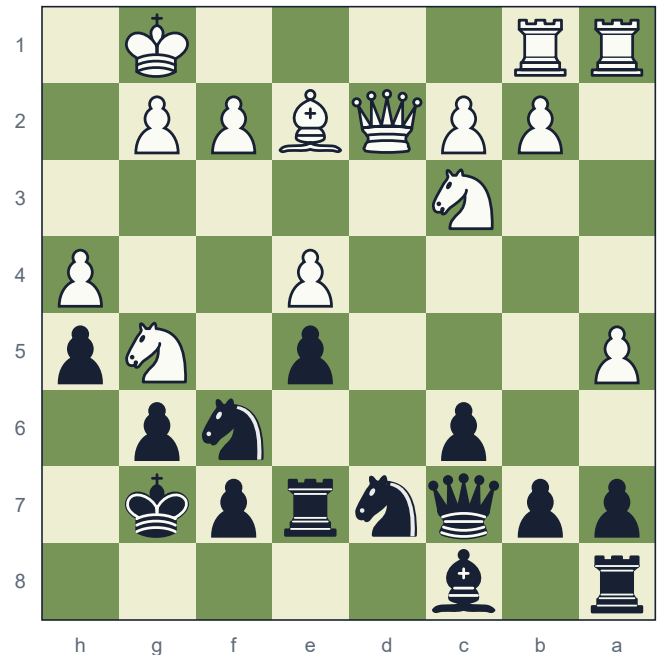
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-03 / play and explain 1

11... e5 12. Bxg7 Kxg7 13. Ng5 Re7 14. dxe5 dxe5 15. Rfb1 Qc7 16. a5



Before this segment - Black to move



After this segment - Black to move

11... E5

Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

13... RE7

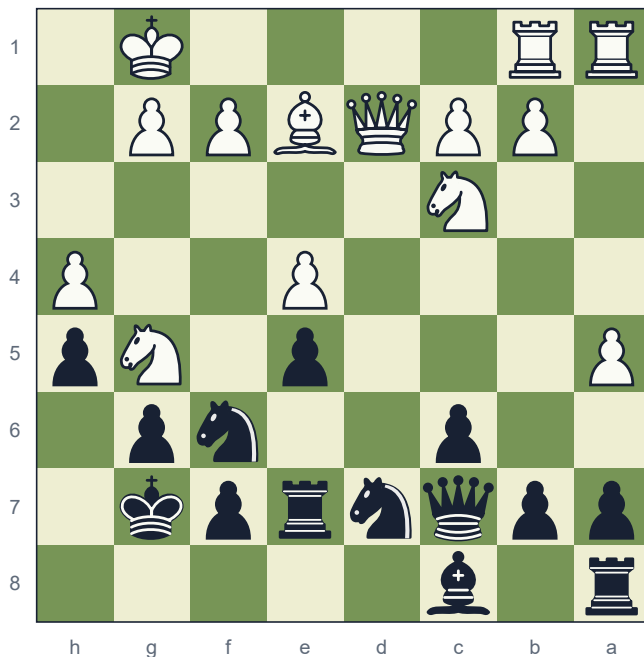
Places the rook on e7. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

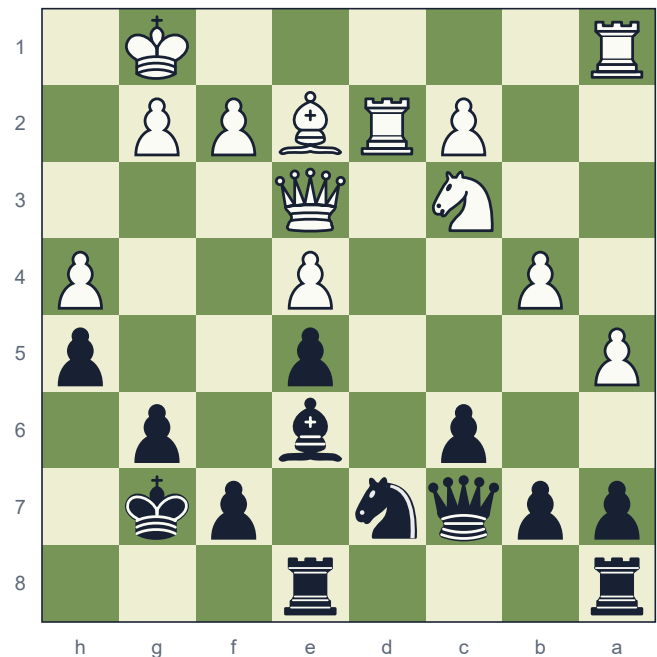
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-03 / play and explain 2

16... Nf8 17. b4 Ne6 18. Nxe6+ Bxe6 19. Qe3 Nd7 20. Rd1 Ree8 21. Rd2



Before this segment - Black to move



After this segment - Black to move

16... NF8

Places the knight on f8. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

18... BXE6

The bishop captures a knight on e6. Calculate the best reply before assuming a recapture, then recount material and opened lines.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-03 / decision 1



Black to move

COMPARE CANDIDATES

e5 and Nf8. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king g8; central-file pawns c6, d6, e7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE E5

11... e5 Search estimate: -0.59 from Black's perspective.

CANDIDATE NF8

11... Nf8 12. Bxg7 Kxg7 13. Ng5 e5 14. f4 exd4 15. Qxd4 Search estimate: -0.64 from Black's perspective.

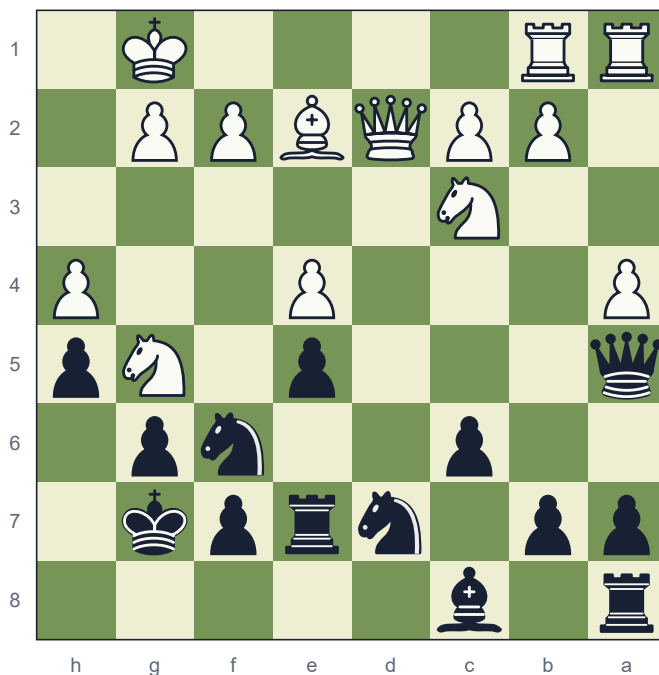
EXPLAIN THE PREFERRED MOVE

Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-03 / decision 2



Black to move

COMPARE CANDIDATES

Qc7 and Nf8. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, e4. Black king g7; central-file pawns e5, c6, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE QC7

15... Qc7 Search estimate: -0.25 from Black's perspective.

CANDIDATE NF8

15... Nf8 16. b4 Search estimate: -0.27 from Black's perspective.

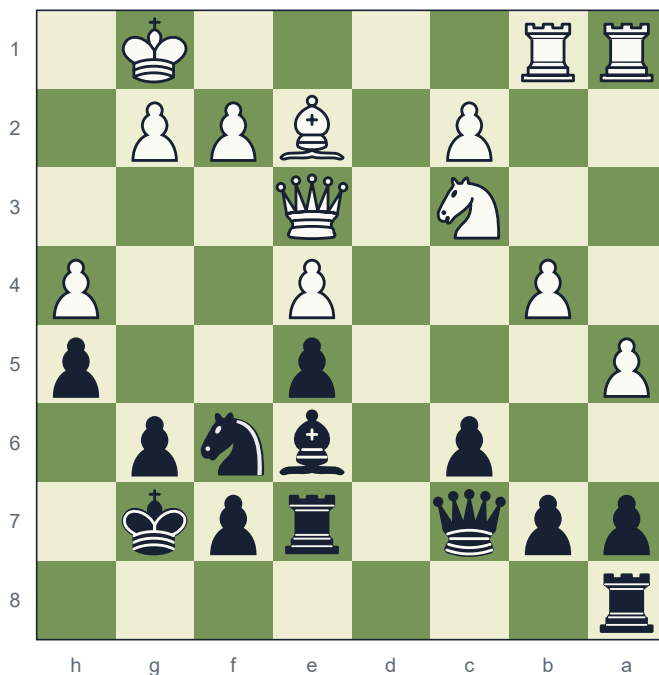
EXPLAIN THE PREFERRED MOVE

Places the queen on c7. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-03 / decision 3



Black to move

COMPARE CANDIDATES

Nd7 and Ree8. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, e4. Black king g7; central-file pawns e5, c6, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE ND7

19... Nd7 20. Rd1 Ree8 21. Rd2 Nf6 22. Rad1 Qe7 23. Rd6 Search estimate: -0.27 from Black's perspective.

CANDIDATE REE8

19... Ree8 20. Na4 Search estimate: -0.35 from Black's perspective.

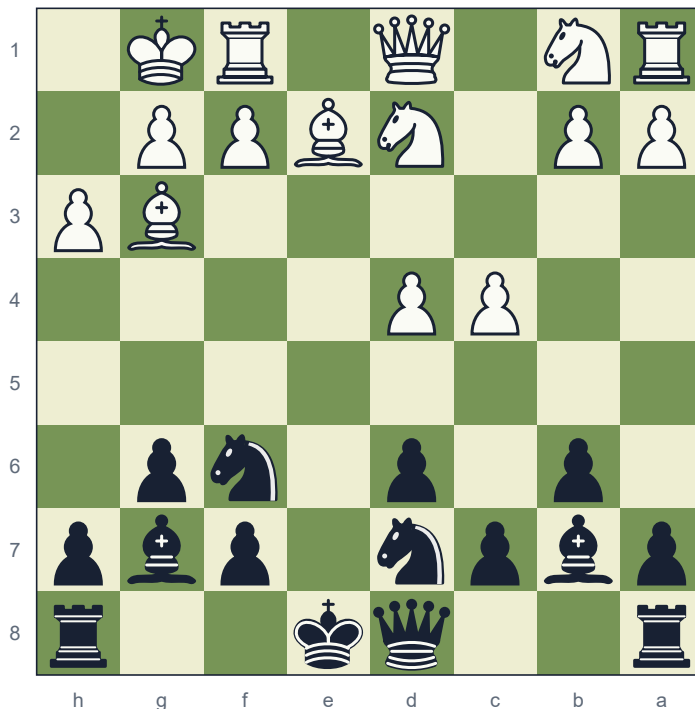
EXPLAIN THE PREFERRED MOVE

Develops a knight toward central supporting squares. Keep a useful onward route so the compact setup does not become overcrowded.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-04 / Against the London bishop



Laboratory start - Black to move

OPENING BACKGROUND

White's bishop develops outside the pawn chain and may support e5. Black uses the double fianchetto while keeping the central break flexible.

READ THE CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king e8; central-file pawns d6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

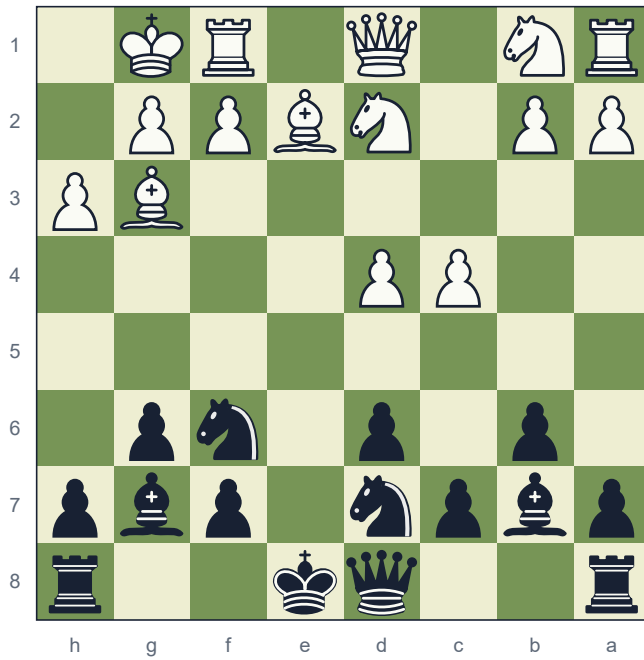
White can use Nbd2, c3 and e4 to expand. The London label does not guarantee that White will remain passive. Decide which parts still apply after the moves already played.

BEFORE PLAYING

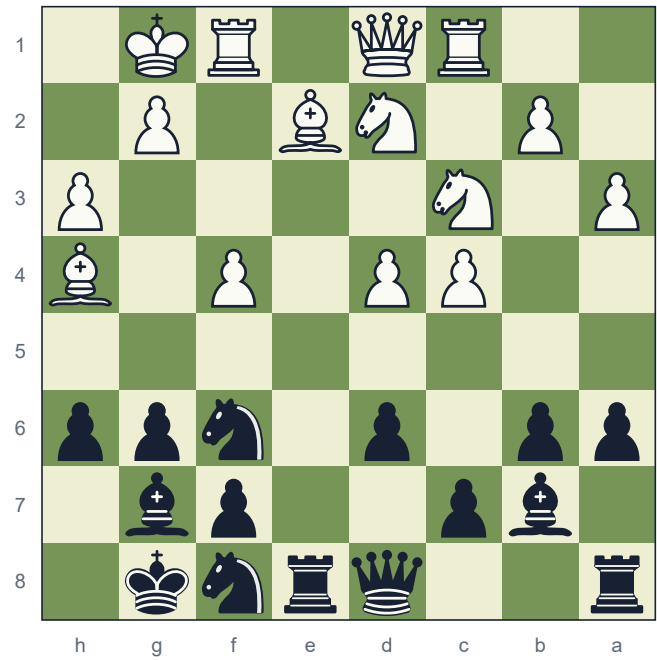
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-04 / play and explain 1

11... O-O 12. Nc3 Re8 13. Bh4 h6 14. Rc1 a6 15. a3 Nf8 16. f4



Before this segment - Black to move



After this segment - Black to move

11... O-O

Moves the king toward shelter and brings a rook toward the center. Check the resulting open lines before assuming the king is safe.

13... H6

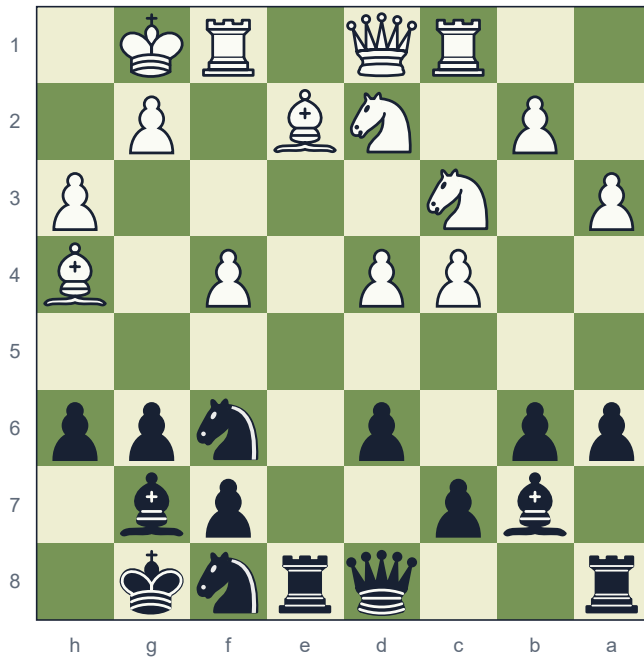
Controls nearby wing squares and may restrain a piece or pawn. Compare this useful purpose with the time lost to central development.

REASSESS

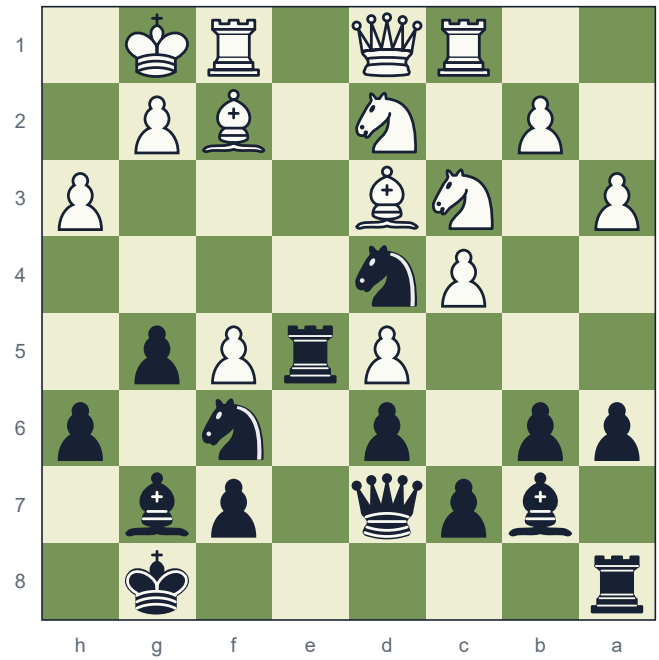
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-04 / play and explain 2

16... Ne6 17. d5 Nd4 18. Bd3 Qd7 19. f5 g5 20. Be1 Re5 21. Bf2



Before this segment - Black to move



After this segment - Black to move

16... NE6

Places the knight on e6. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

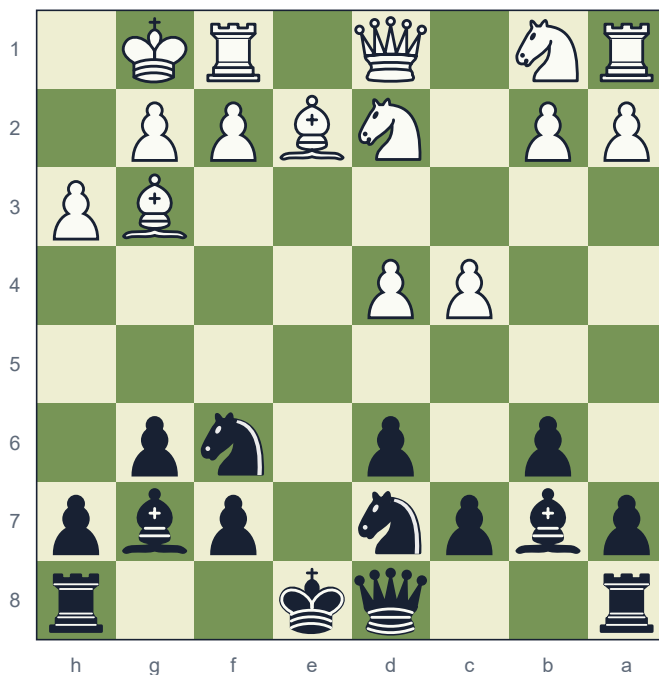
18... QD7

Places the queen on d7. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-04 / decision 1



Black to move

COMPARE CANDIDATES

O-O and h6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king e8; central-file pawns d6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE O-O

11... O-O 12. Nc3 Re8 13. Bh4 h6 14. a4 a6 15. Rc1 Search estimate: -0.47 from Black's perspective.

CANDIDATE H6

11... h6 12. Nc3 O-O 13. Qc2 Nh7 14. Bf3 Bxf3 15. Nxf3 Search estimate: -0.48 from Black's perspective.

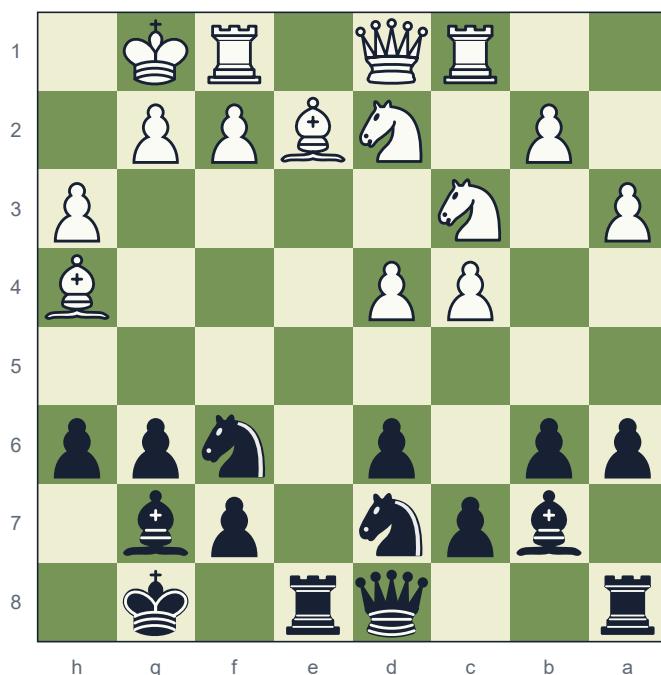
EXPLAIN THE PREFERRED MOVE

Moves the king toward shelter and brings a rook toward the center. Check the resulting open lines before assuming the king is safe.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-04 / decision 2



Black to move

COMPARE CANDIDATES

Nf8 and Qc8. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c4, d4. Black king g8; central-file pawns d6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NF8

15... Nf8 16. f4 Ne6 17. d5 Nd4 18. Bd3 Bc8 19. Nb3 Search estimate: -0.51 from Black's perspective.

CANDIDATE QC8

15... Qc8 16. Re1 Search estimate: -0.54 from Black's perspective.

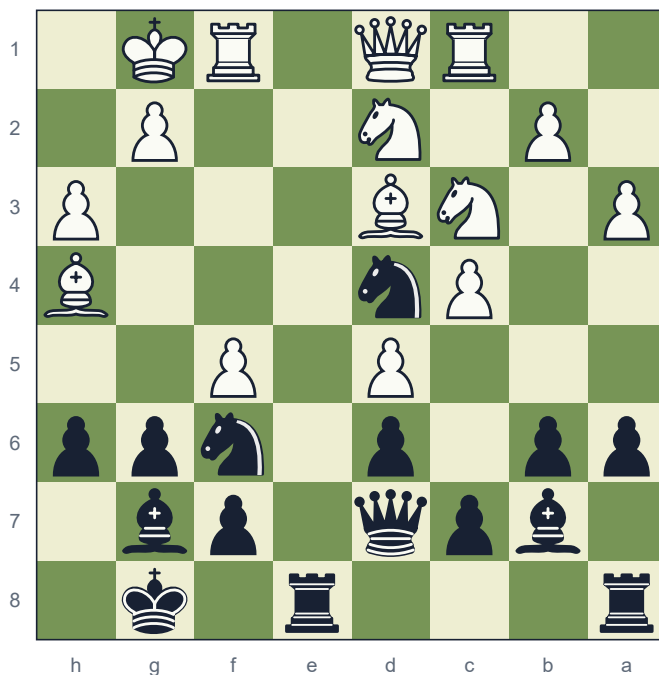
EXPLAIN THE PREFERRED MOVE

Places the knight on f8. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-04 / decision 3



Black to move

COMPARE CANDIDATES

g5 and Nxf5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c4, d5, f5. Black king g8; central-file pawns d6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE G5

19... g5 Search estimate: -0.46 from Black's perspective.

CANDIDATE NXF5

19... Nxf5 20. Bxf5 gxf5 21. Rc2 b5 22. Nf3 bxc4 23. Nd4 Search estimate: -0.70 from Black's perspective.

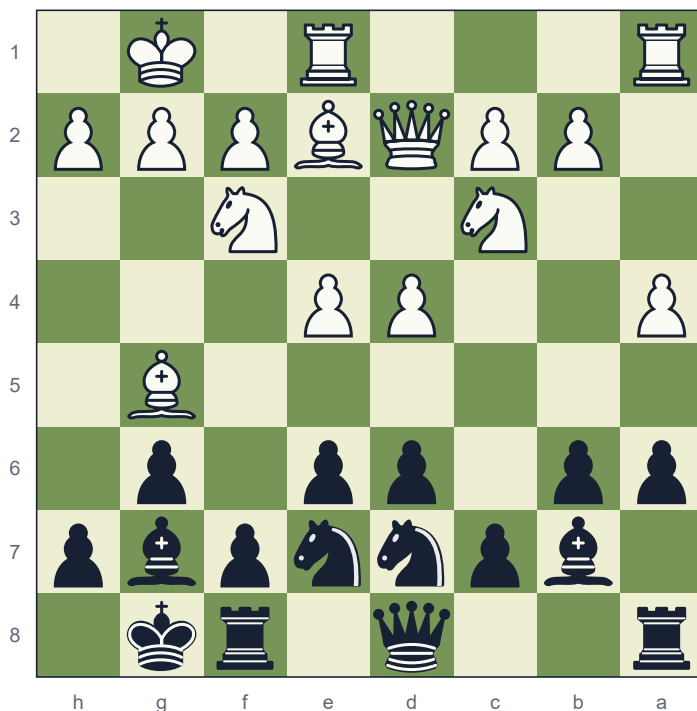
EXPLAIN THE PREFERRED MOVE

Places the pawn on g5, attacking bishop on h4. A legal capture may still be unsound if the target is defended.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-05 / Prepare ...c5 with development



Laboratory start - Black to move

OPENING BACKGROUND

Black develops the queen knight before forcing a central break. In this exact position an immediate ...c5 allows strong white play; a thematic move still needs preparation.

READ THE CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king g8; central-file pawns d6, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

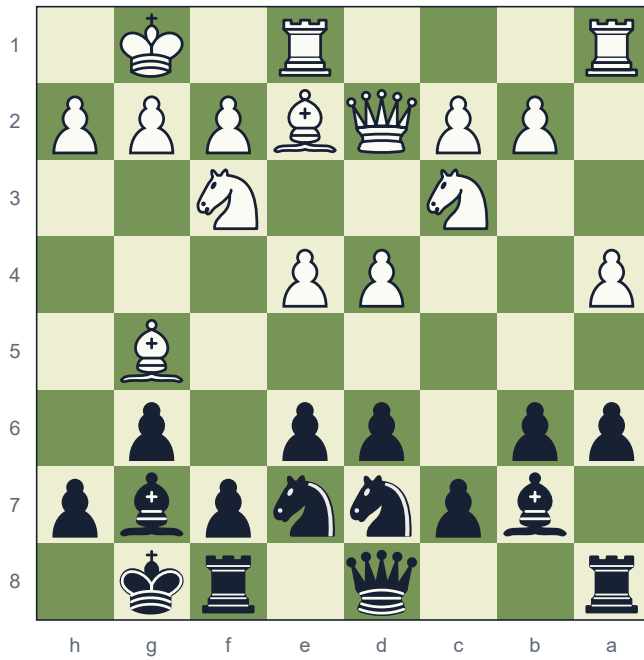
White can improve with Qd2 or advance in the center. Do not assume that a break becomes good simply because it belongs to the opening's vocabulary. Decide which parts still apply after the moves already played.

BEFORE PLAYING

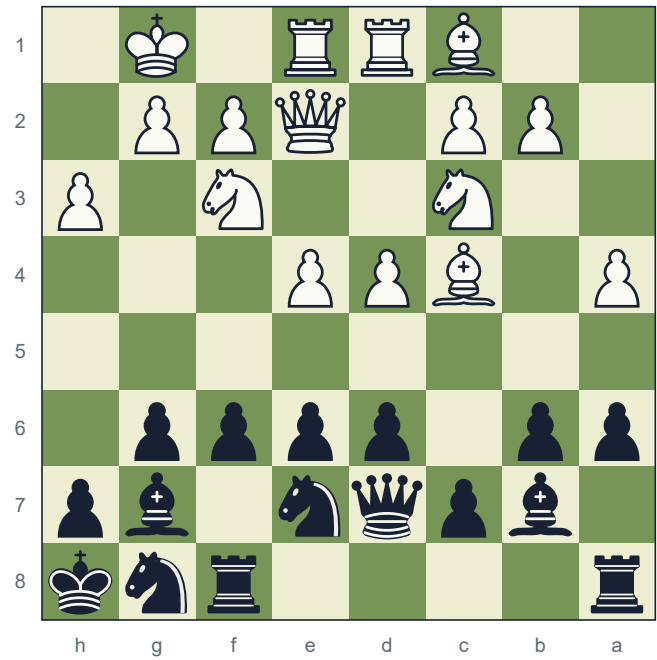
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-05 / play and explain 1

11... Kh8 12. Bc4 Nf6 13. Rad1 Qd7 14. h3 Nfg8 15. Qe2 f6 16. Bc1



Before this segment - Black to move



After this segment - Black to move

11... KH8

Places the king on h8. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

13... QD7

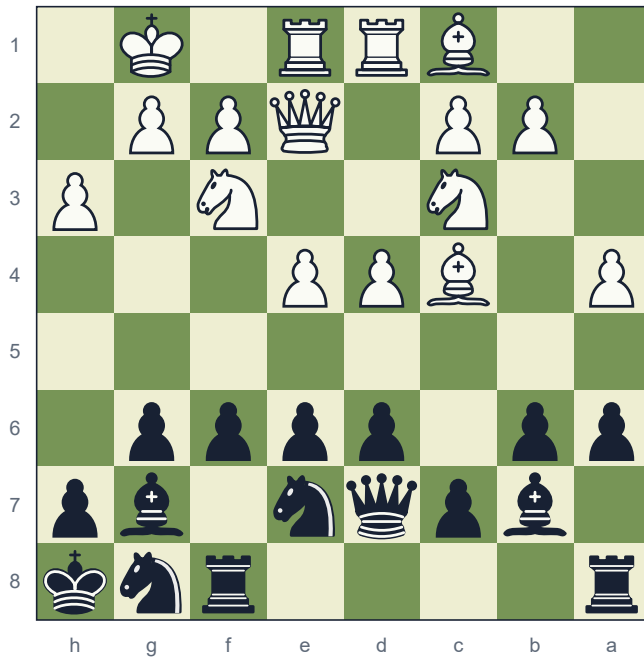
Places the queen on d7. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

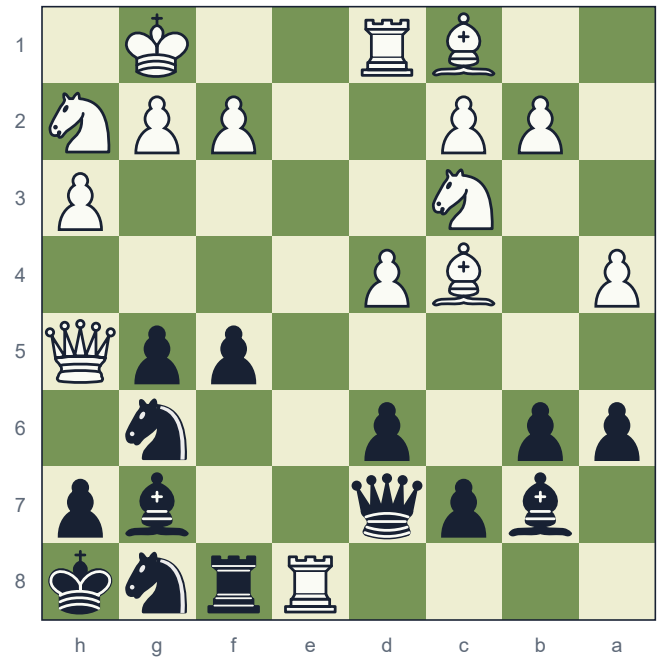
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-05 / play and explain 2

16... g5 17. Nh2 Ng6 18. Qg4 f5 19. exf5 exf5 20. Qh5 Rae8 21. Rxe8



Before this segment - Black to move



After this segment - Black to move

16... G5

Creates a wing or central pawn contact and changes king shelter. Calculate captures and the opponent's central counterplay before extending the advance.

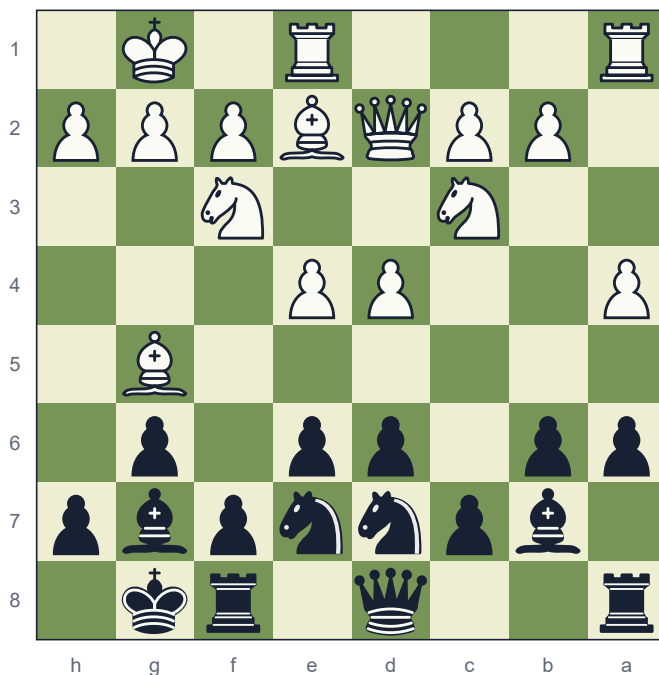
18... F5

Places the pawn on f5, attacking queen on g4. A legal capture may still be unsound if the target is defended.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-05 / decision 1



Black to move

COMPARE CANDIDATES

Kh8 and Re8. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king g8; central-file pawns d6, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE KH8

11... Kh8 Search estimate: -0.76 from Black's perspective.

CANDIDATE RE8

11... Re8 12. h4 Nf6 13. Bd3 h5 14. e5 Nh7 15. Ne4 Search estimate: -0.79 from Black's perspective.

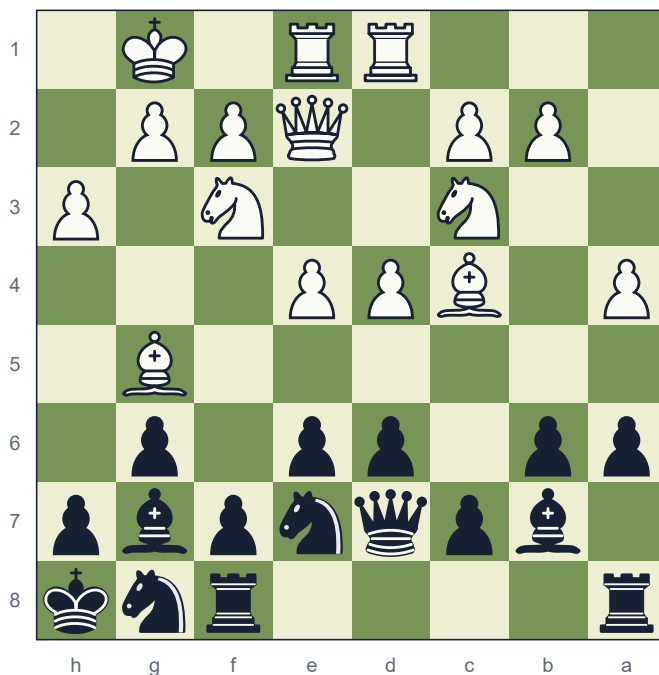
EXPLAIN THE PREFERRED MOVE

Places the king on h8. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-05 / decision 2



Black to move

COMPARE CANDIDATES

f6 and h6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, e4. Black king h8; central-file pawns d6, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE F6

15... f6 16. Bc1 g5 17. Bb3 Ng6 18. Qd3 Qf7 19. Ne2 Search estimate: -0.70 from Black's perspective.

CANDIDATE H6

15... h6 16. Bf4 Search estimate: -0.81 from Black's perspective.

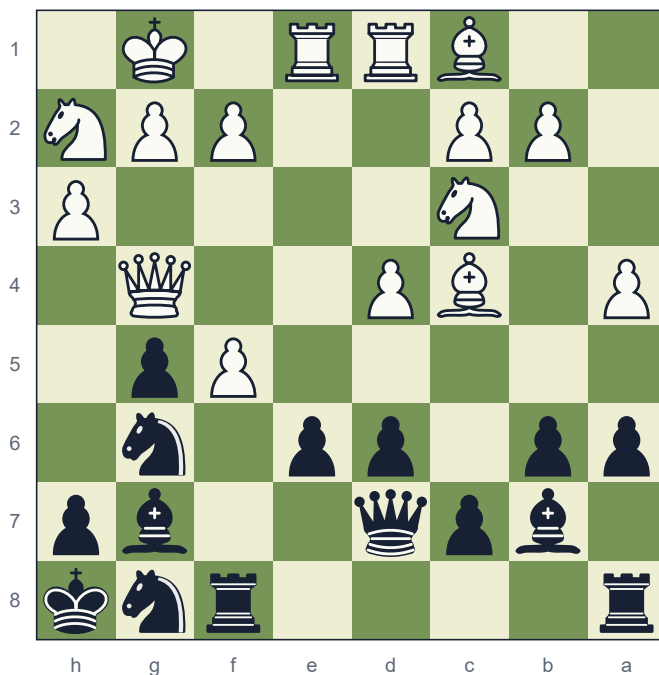
EXPLAIN THE PREFERRED MOVE

Places the pawn on f6, attacking bishop on g5. A legal capture may still be unsound if the target is defended.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-05 / decision 3



Black to move

COMPARE CANDIDATES

exf5 and Nh6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d4, f5. Black king h8; central-file pawns d6, e6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE EXF5

19... exf5 20. Qh5 Search estimate: -0.95 from Black's perspective.

CANDIDATE NH6

19... Nh6 20. Bxe6 Qd8 21. Qh5 Nxf5 22. Bxg5 Nxd4 23. Bxd8 Search estimate: -2.76 from Black's perspective.

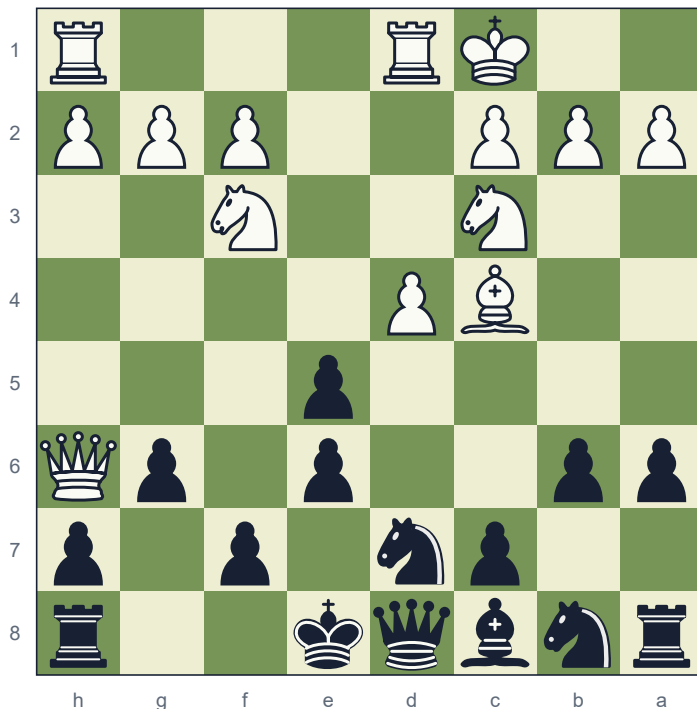
EXPLAIN THE PREFERRED MOVE

The pawn captures a pawn on f5. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-06 / White tries Bh6



Laboratory start - Black to move

OPENING BACKGROUND

Black develops Nf6 before this bishop-exchange sequence. White removes the fianchetto bishop and places the queen near the king side, creating a real dark-square concern.

READ THE CURRENT POSITION

White king c1; central-file pawns c2, f2, d4. Black king e8; central-file pawns e5, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

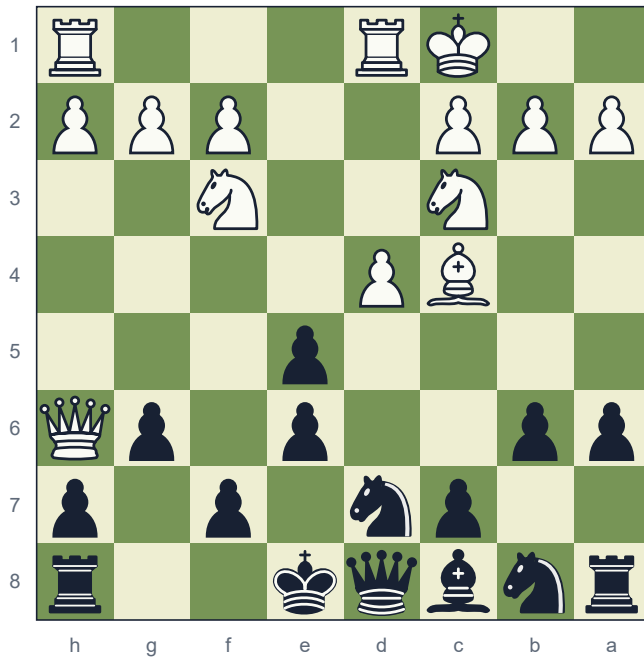
If the black knight were still on g8, Qxh6 would lose the queen to ...Nxh6. The developed knight is an essential detail in this line. Decide which parts still apply after the moves already played.

BEFORE PLAYING

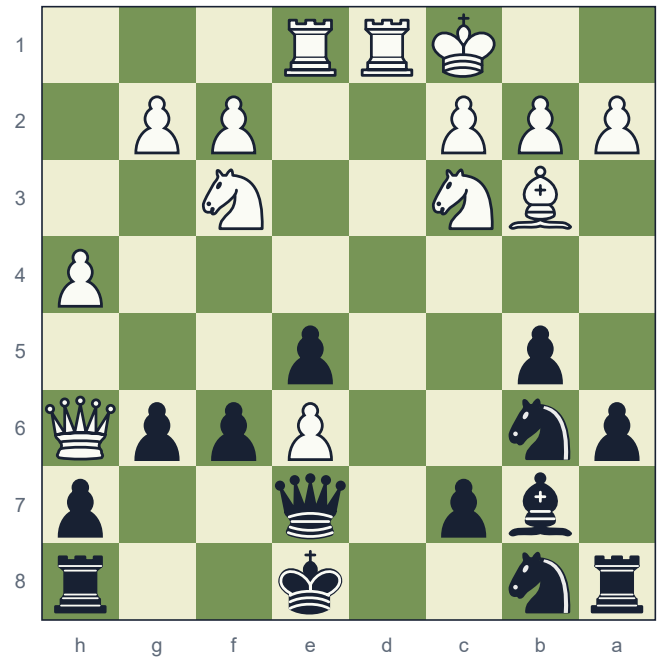
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HB-LAB-06 / play and explain 1

11... Bb7 12. d5 Qe7 13. Rhe1 b5 14. Bb3 f6 15. dxe6 Nb6 16. h4



Before this segment - Black to move



After this segment - Black to move

11... BB7

Places the bishop on b7, attacking knight on f3. A legal capture may still be unsound if the target is defended.

13... B5

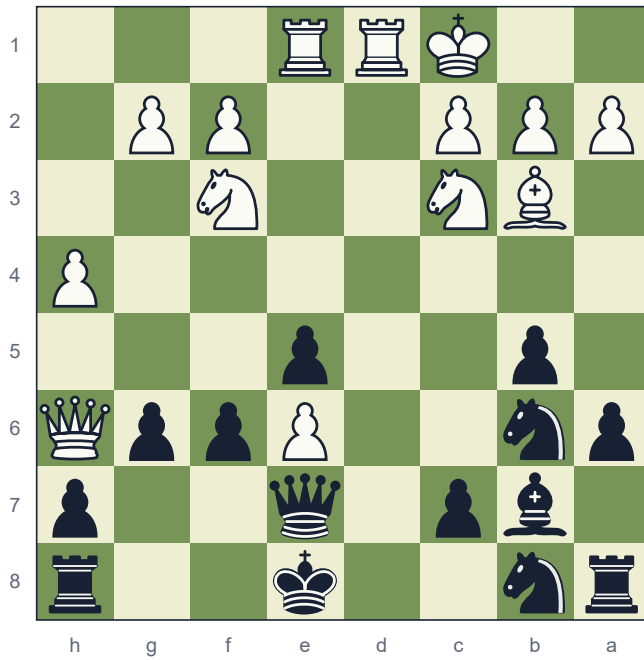
Places the pawn on b5, attacking bishop on c4. A legal capture may still be unsound if the target is defended.

REASSESS

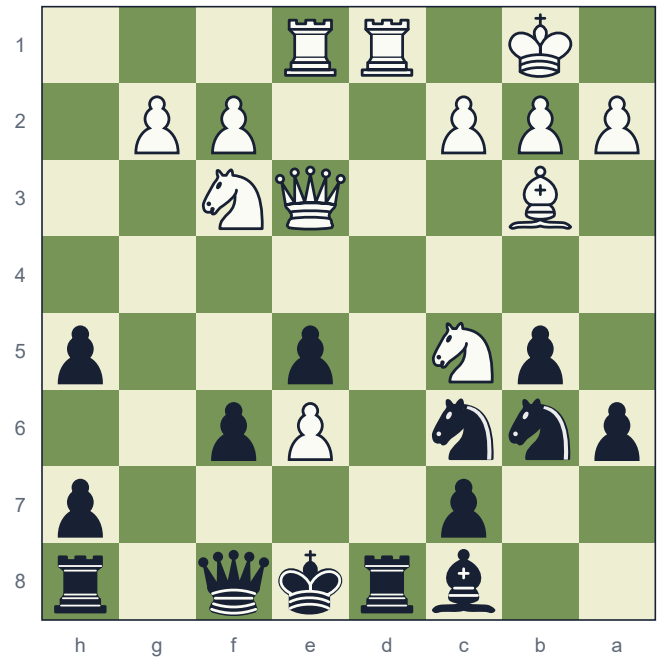
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-06 / play and explain 2

16... Nc6 17. h5 Qf8 18. Qe3 Rd8 19. Ne4 gxh5 20. Nc5 Bc8 21. Kb1



Before this segment - Black to move



After this segment - Black to move

16... NC6

Develops toward d4 and e5. Check whether an opposing d5 advance would gain a tempo.

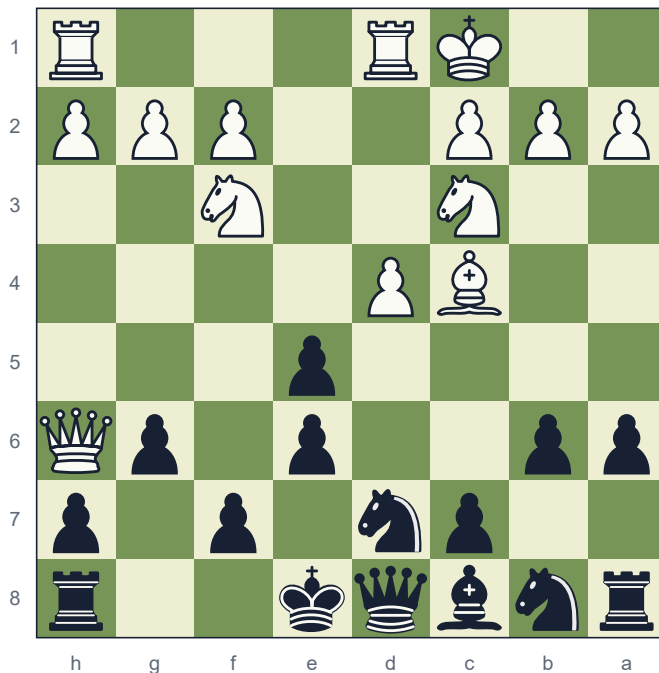
18... RD8

Places the rook on d8, attacking rook on d1. A legal capture may still be unsound if the target is defended.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HB-LAB-06 / decision 1



Black to move

COMPARE CANDIDATES

Bb7 and b5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king c1; central-file pawns c2, f2, d4. Black king e8; central-file pawns e5, e6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BB7

11... Bb7 12. d5 Qe7 13. Rhe1 b5 14. Bb3 f6 15. dxe6 Search estimate: -2.09 from Black's perspective.

CANDIDATE B5

11... b5 12. Bb3 Qe7 13. d5 Qf8 14. Qh3 Nc5 15. Nxe5 Search estimate: -2.14 from Black's perspective.

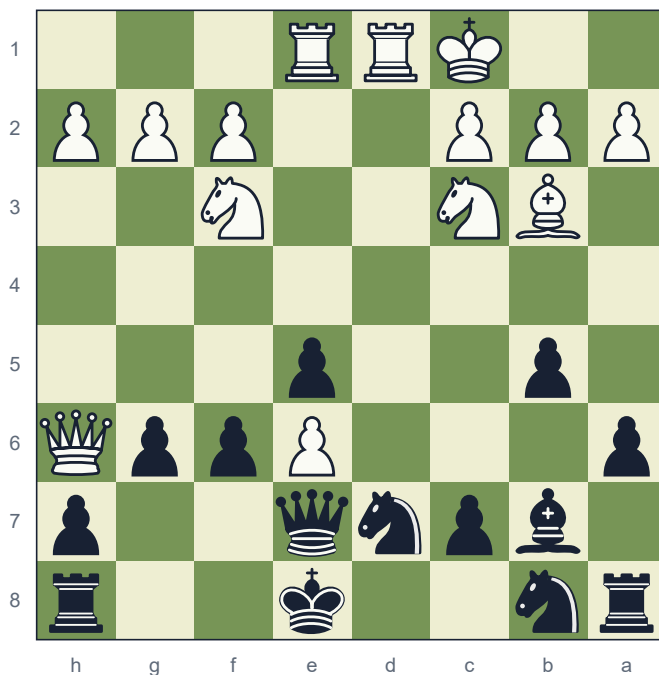
EXPLAIN THE PREFERRED MOVE

Places the bishop on b7, attacking knight on f3. A legal capture may still be unsound if the target is defended.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-06 / decision 2



Black to move

COMPARE CANDIDATES

Nb6 and Nc5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king c1; central-file pawns c2, f2, e6. Black king e8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NB6

15... Nb6 16. h4 Nc6 17. h5 Qf8 18. Qe3 Rd8 19. Ne4 Search estimate: -2.49 from Black's perspective.

CANDIDATE NC5

15... Nc5 16. Nd2 Nxb3+ 17. Nxb3 b4 18. Nd5 Bxd5 19. Rxd5 Search estimate: -2.64 from Black's perspective.

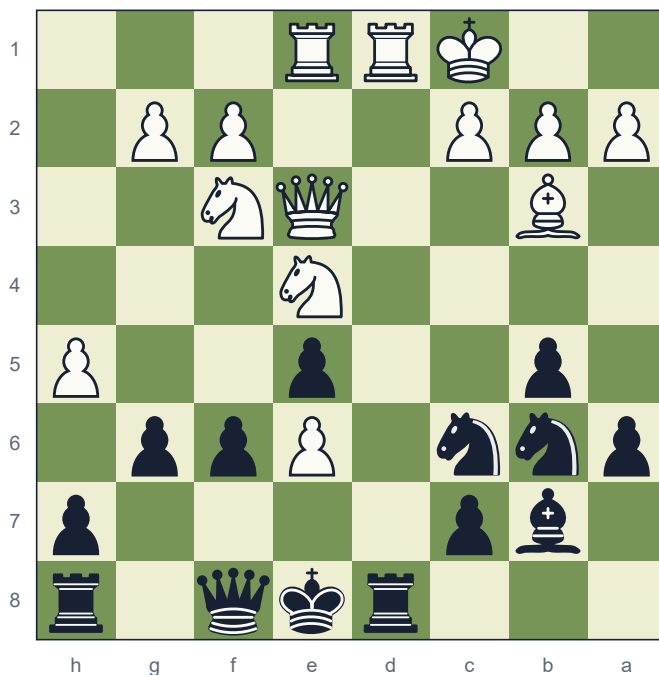
EXPLAIN THE PREFERRED MOVE

Places the knight on b6. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HB-LAB-06 / decision 3



Black to move

COMPARE CANDIDATES

gxh5 and Qg7. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king c1; central-file pawns c2, f2, e6. Black king e8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE GXH5

19... gxh5 20. Nc5 Rxd1+ 21. Rxd1 Bc8 22. Kb1 Nc4 23. Bxc4 Search estimate: -2.44 from Black's perspective.

CANDIDATE QG7

19... Qg7 20. Nc5 Rxd1+ 21. Rxd1 Bc8 22. Kb1 Na5 23. Nh4 Search estimate: -2.73 from Black's perspective.

EXPLAIN THE PREFERRED MOVE

The pawn captures a pawn on h5. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.